Deep Reinforcement Learning

RL + DL

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Q-learning → Q-table $Q(s,a)$ ⟹ 48 states
(discrete, small)

↓

Deep Q-Networks (DQN)

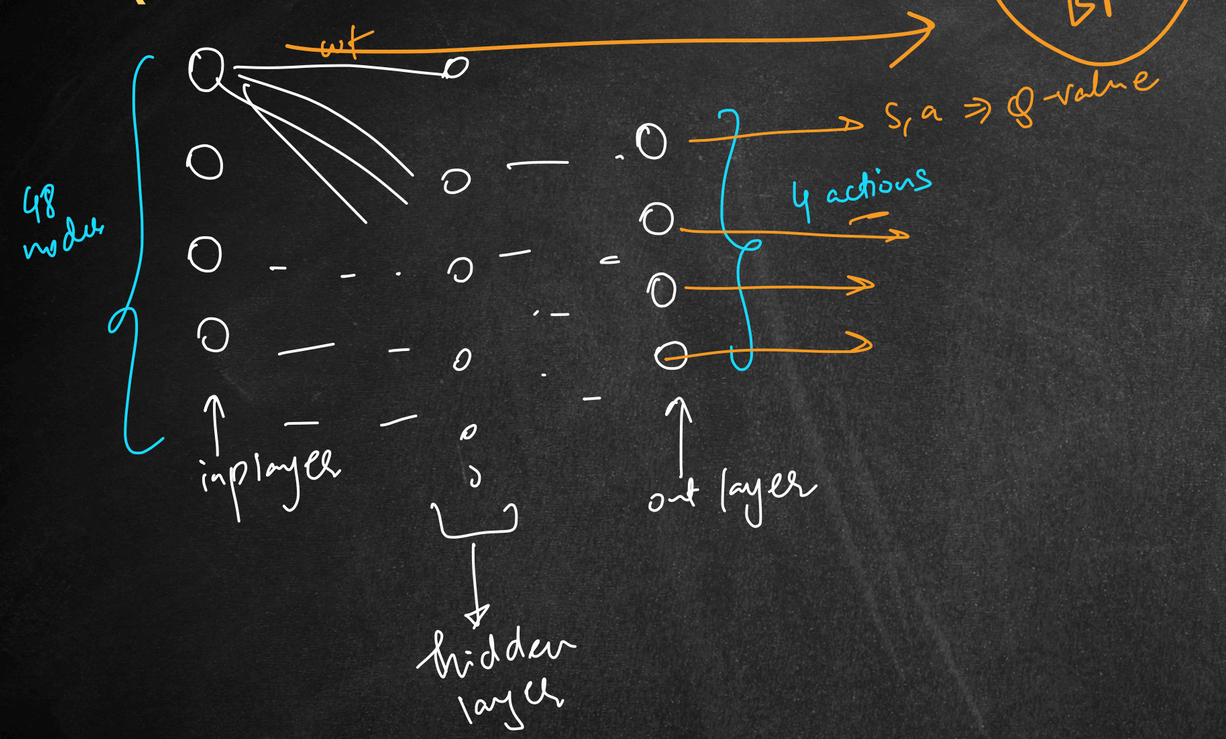
problem
1. millions of states → huge
2. continuous state val → impossible

state → NN function → actions
Q-vals

approximate or predict

Deep Q-Networks

DQN



48 × 4 ⟶ Q-table
states action
Q-val

Q-learning ⇒ Q(s, a)

DQN ⇒ Q(s, a; θ)

NN parameters

① Correlated samples

② Moving Target

Deep Q-Networks

Correlated Samples & Experience Replay

Problem

NN → independent

E1: S1 → A1 → R2 → S2

E2: S2 → A2 → R3 → S3

E3: S3 → A3 → R4 → S4

E1 → E2 → E3 ----

Naive DQN

experiences = State, Action, Reward, Next State, Terminal

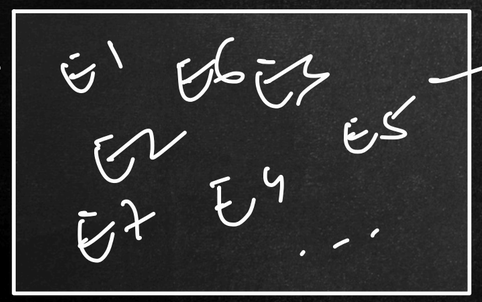
1. sequential
2. correlated

batch size = 16

✗ E1 → E2 → E3 → E4 ----

✓ E9 → E121 → E2 → E5

Solⁿ: ER ⇒ DQN memory (replay buffer)



randomly sample minibatch of Ex

Deep Q-Networks

Moving Target & Target Network

leads to → unstable training

Q-learning update ⇒ $Q(s,a) = Q(s,a) + \alpha\left[r + \gamma \cdot \max_a Q(s',a) - Q(s,a)\right]$

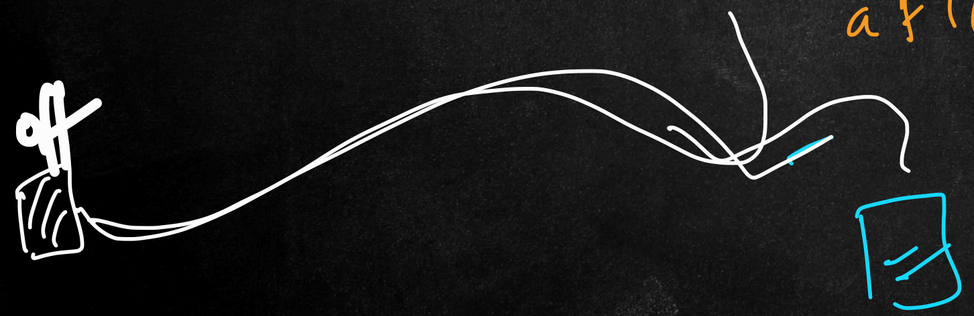
target ⇒ $y = r + \gamma \cdot \max_a Q(s', a; \theta)$ → y

predict Q-value $= Q(s', a; \theta)$ → $\hat{y}$

loss $= (y - \hat{y})$

$=$

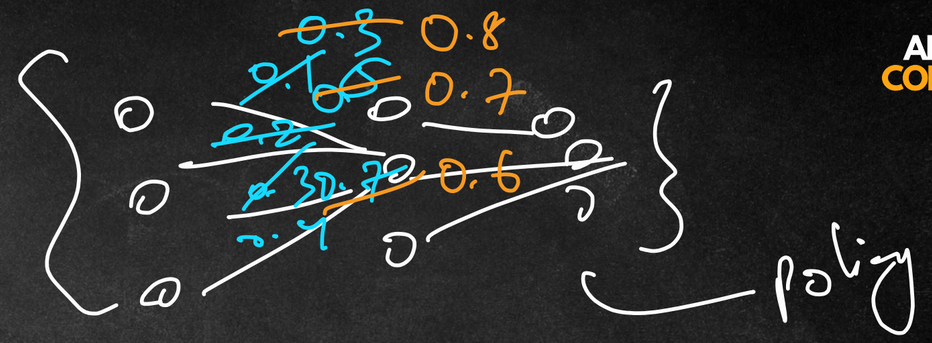
target is not a FIXED value



Deep Q-Networks

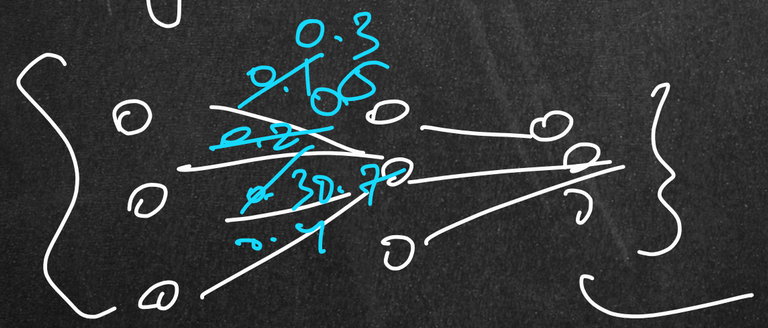
Moving Target & Target Network

Naive DQN ⟶ 1 NN



DQN

1. Policy Network (online)
2. Target Network (old copy of policy network)



architecture

single

Deep Q-Networks

Training Flappy Bird

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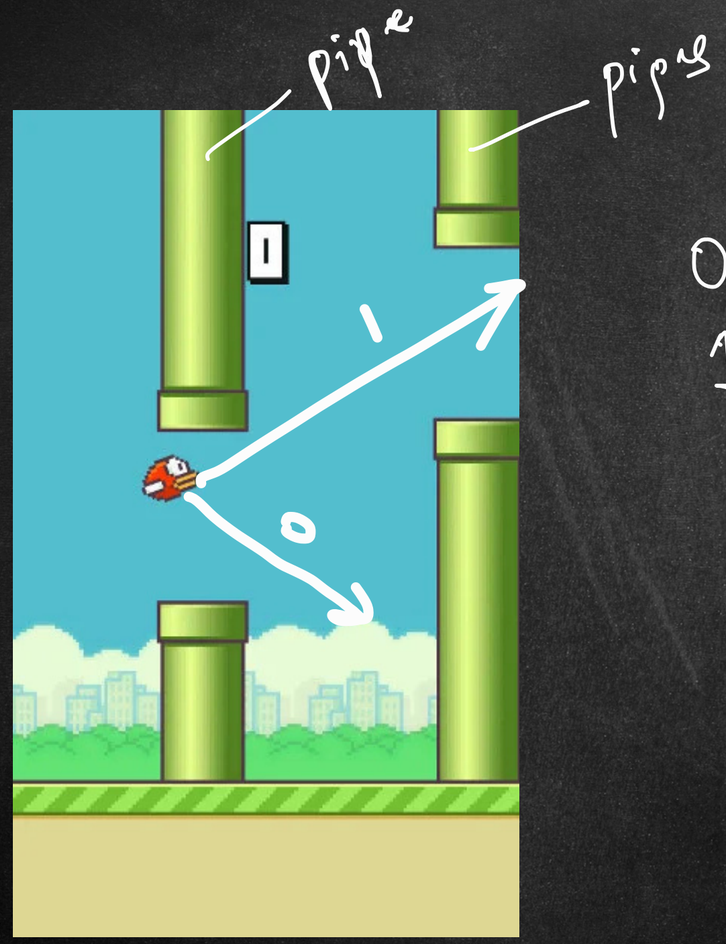
Deep Q-Networks

Training Flappy Bird

Agent

Reinforcement learning
Minor Project

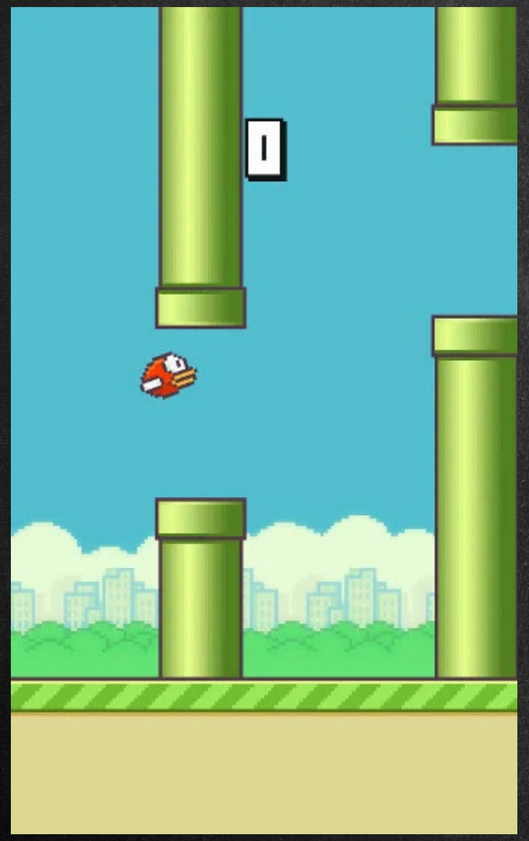


0 → do nothing
1 → flap

Deep Q-Networks

Training Flappy Bird

LIDAR



$\epsilon = 1$
↓
0.9
↓
0.8
↓
0.7
⋮
↓
0.05

0. Set Env
1. Define DQN
2. Create the Experience Replay
3. multiple episodes DQN train → setup
4. Epsilon-greedy policy + hyperparams (epsilon decay)
7. Target Network
8. Train & Test

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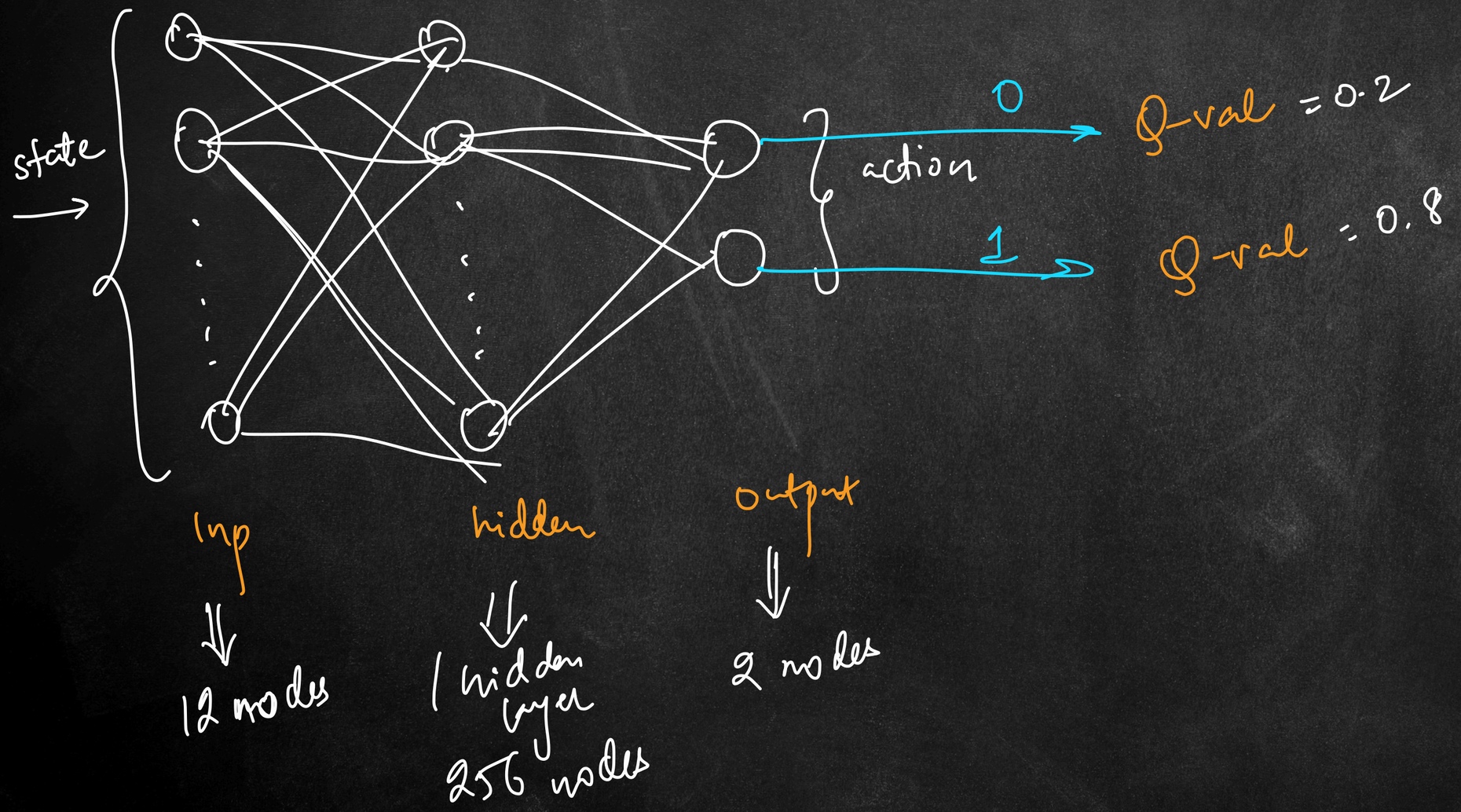
Deep Q-Networks

Training Flappy Bird → DQN → neural network arch



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Deep Q-Networks

Training Flappy Bird

① device → CPU
GPU → Apple (m1, m2, m3) ⇒ mps
GPU → nvidia ⇒ CUDA

② run()
Train
Test & Visualize/Render

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Deep Q-Networks

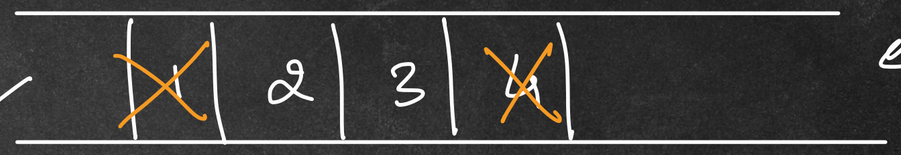
Training Flappy Bird

↓

Experience Replay

Deque (double ended queue)

add/rem



add/remove

add: 1, 2, 3, 4

remove

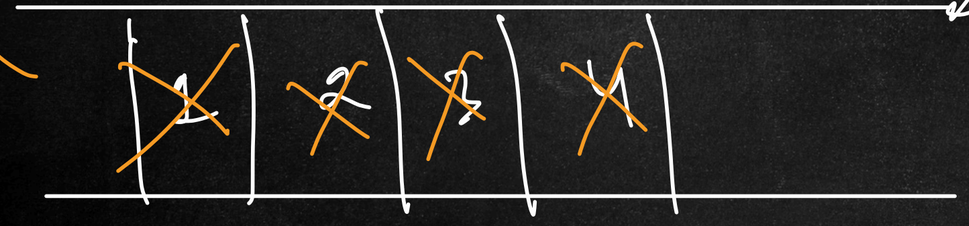
FIFO mode — First In First Out

LIFO — Last In First Out

→ stack

remove

rear end add

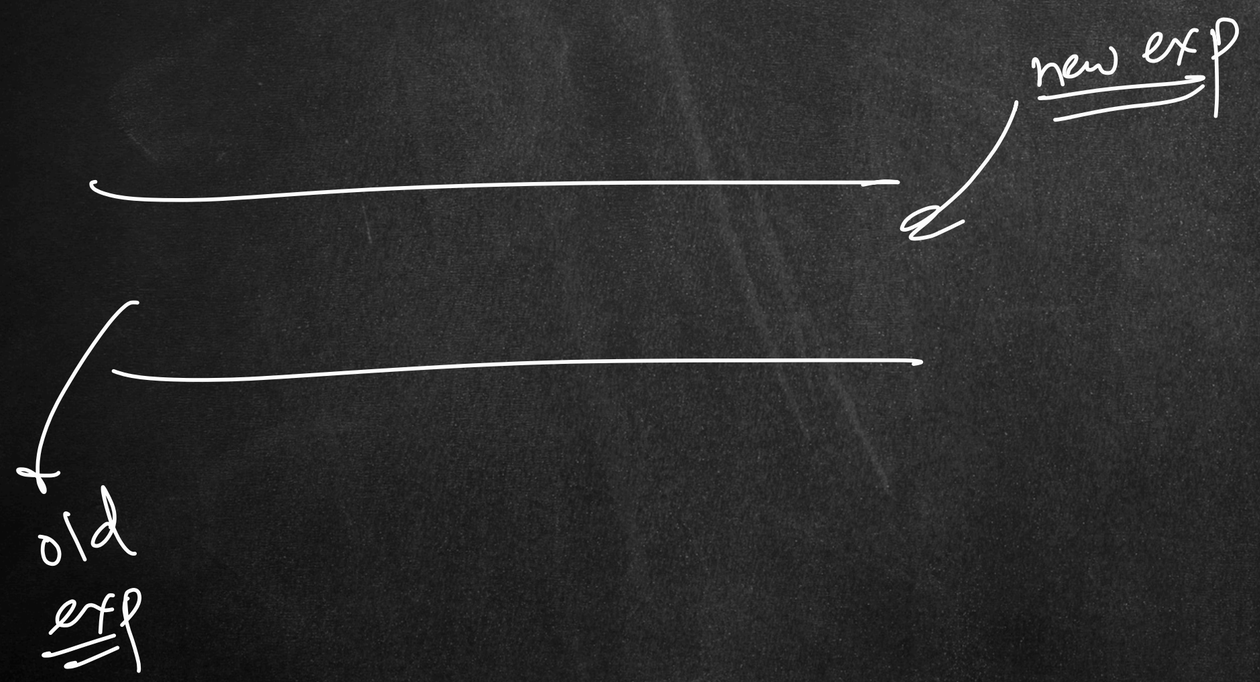


1 2 3 4

Deep Q-Networks

Training Flappy Bird

experience : State, Action, Reward, Next State, Terminated



Deep Q-Networks

Training Flappy Bird

2000 ep

5000 ep

dqn θ

Policy → 10 steps

Target

indefinitely

1. manually stop (ctrl+c)
2. reward_threshold → stop

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Deep Q-Networks

Training Flappy Bird

→ ε- greedy policy

→ ε- decay

→ tensors

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